

The Central Distribution: Representing Hierarchical Partial Knowledge

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Abstract

We introduce the *central distribution*, a probability distribution that represents a set or hierarchy of probability distributions by minimizing the worst-case Kullback–Leibler (KL) divergence to the admissible models. This provides a principled method for converting partial knowledge and set-valued uncertainty into a single probability measure suitable for Bayesian inference.

For finite sets of distributions, we show that central distributions admit a minimax characterization and can be obtained as solutions to convex–concave saddle-point problems. We establish existence, uniqueness, and characterization results. For parametric models, we derive a sufficient fixed-point condition for priors whose induced marginals are central, revealing a formal connection with Bernardo’s reference prior methodology.

We extend the framework to hierarchical uncertainty structures and propose an asymptotic notion of centrality for infinite sequences. We further derive approximation bounds that lead to practical procedures for constructing approximate central distributions and performing Bayesian inference without explicit knowledge of the exact solution.

Finally, we illustrate how uncertainty about relevant feature subsets can be represented hierarchically and aggregated into a single predictive model. These results suggest a general information-theoretic framework for representing partial knowledge, constructing noninformative priors, and reasoning under model uncertainty.

1 Introduction

Consider a standard decision problem in Bayesian decision theory. An agent must choose an action from a set of available alternatives. The resulting loss depends on an unknown state of the environment and is quantified by a real-valued loss function. Specifically, let a denote an action and x the unknown state. The loss associated with choosing action a when the state is x is given by the loss function $\ell(a, x)$.

We assume that some evidence e about the unknown state is available and can be used to guide the agent's decision. In other words, the agent aims to find an optimal strategy, where a strategy is a conditional distribution over actions given the observed evidence, denoted by $S(A | E)$.

In Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg \min_S \sum_a \sum_{x,e} \ell(a, x) S(a | e) P(x, e).$$

One can show that there always exists a deterministic optimal strategy, namely a strategy that, for each evidence e , selects the action minimizing the expected loss conditioned on e :

$$S_{\text{opt}}(e) = \arg \min_a \mathbb{E}_P [\ell(a, x) | e].$$

Now suppose that, instead of basing our decisions on the original evidence E , we choose to use a transformed version E' . In practice, the evidence available to the agent is fixed by the environment. However, the agent may choose to base its decisions on a deterministic transformation $E' = f(E)$. This is analogous to feature selection in machine learning, where a high-dimensional observation is replaced by a lower-dimensional representation.

Could such a transformation improve the agent's strategy? Within the framework of Bayesian decision theory, this question has a straightforward answer. Any two strategies can be compared through their expected loss: the strategy with the lower expected loss is preferred. However, a fundamental result of Bayesian decision theory states that the optimal strategy based on E is always at least as good as any strategy based on E' .

One might argue that adopting a strategy that uses simpler evidence may discard information and is therefore not justified by Bayesian decision theory. The key point here is that the space of strategies using E' is a subset of the space of strategies using E . When we search for the optimal strategy based on E , we are simultaneously considering all strategies that depend on E' , since every such strategy can also be expressed as a strategy based on

E . This helps explain why standard Bayesian inference does not inherently favor simpler representations of the evidence.

Although the optimal Bayesian strategy based on E can always exploit all available information, the joint distribution $P(x, e)$ determines which aspects of the evidence are actually useful for predicting the unknown state. With an appropriate choice of $P(x, e)$, the optimal strategy based on E may effectively ignore irrelevant components of the evidence and behave as though it were based on a lower-dimensional representation E' . In this sense, Bayesian inference can perform a form of automatic feature selection.

In many statistical models, specifying $P(x, e)$ amounts to choosing a prior distribution. Consequently, the ability to promote relevant information depends on how prior knowledge about the environment is encoded. The difficulty therefore does not lie in Bayesian decision theory itself. Rather, it lies in how partial knowledge is represented. If partial knowledge is represented appropriately, standard Bayesian inference can naturally extract relevant information while disregarding irrelevant aspects of the evidence. Many commonly used priors, including flat priors, do not generally provide this behavior. The central distribution framework developed in the present work is proposed to address this problem by providing a principled representation of partial knowledge for Bayesian inference.

In this paper, we introduce the *central distribution*, an information-theoretic representation of partial knowledge that converts set-valued, and more generally hierarchical, uncertainty into a single probability distribution suitable for Bayesian inference. We establish the theoretical foundations of the framework by developing the theory of central distributions for finite sets of probability distributions, including existence, uniqueness, and minimax characterizations, and by introducing the corresponding notion of central priors for parametric models. We further identify a sufficient fixed-point condition linking central priors to objective Bayesian reference analysis [2]. Building on these foundations, we investigate asymptotic extensions of the framework and develop a practical approximation method for constructing central distributions. Finally, we illustrate how representing uncertainty about model structure through hierarchical central distributions allows standard Bayesian inference to naturally perform feature selection.

1.1 Related Work

The problem of constructing objective or noninformative priors has a long history in Bayesian statistics. Among the most influential approaches are Jeffreys' priors, reference priors, and maximum-entropy based methods [5, 1, 4].

Jeffreys introduced a general rule for constructing invariant noninformative priors based on the Fisher information matrix [5]. The resulting priors are widely used because of their parameterization invariance and their connection to asymptotic likelihood theory. However, Jeffreys’ priors are derived from local geometric properties of the parameter space and do not directly address the problem of representing a set of admissible probability distributions by a single distribution.

The closest connection to the present work is Bernardo’s reference prior methodology [2, 1]. Reference priors are constructed by maximizing the expected information gained from the data and are intended to represent a state of minimal prior information. In Section 2.2, we show that central priors satisfy a fixed-point condition closely related to the reference prior construction. In particular, for a single parameter and a single set of distributions, the resulting central prior coincides with the corresponding reference prior.

Several authors have also studied objective Bayesian inference through information-theoretic principles [4, 7]. Examples include maximum entropy methods, minimum discrimination information principles, and related divergence based approaches. These methods typically select a distribution by optimizing an information criterion subject to constraints. In contrast, the central distribution is defined as a minimax representative of a set or hierarchy of distributions, minimizing the worst-case Kullback–Leibler divergence to the admissible models.

The present work is also related to robust Bayesian statistics and imprecise probability [8, 9]. In these frameworks uncertainty is represented by sets of probability distributions rather than a single model. Existing approaches generally focus on performing inference directly with sets of distributions. The central distribution framework instead seeks a single representative distribution that summarizes the available set-valued or hierarchical uncertainty while preserving its information-theoretic structure.

The central distribution framework differs from these approaches in two respects. First, it is defined as a minimax-KL representative of a set, or more generally a hierarchy, of probability distributions. Second, it provides a mechanism for converting set-valued and hierarchical uncertainty into a single probability distribution suitable for Bayesian inference. While related to objective Bayesian inference, robust Bayesian analysis, and information-theoretic methods, its primary goal is not the direct specification of a prior distribution for parameters or inference with sets of distributions, but rather the construction of a representative distribution that preserves the informational structure of the underlying uncertainty.

2 Central Distribution

Consider a random variable X . In Bayesian modeling, our knowledge about X is represented by a probability distribution, which serves as a basis for making inferences.

However, in many situations, our information may be too limited to justify choosing a single distribution. Instead, we may be able to specify a set of plausible distributions. Such a set represents some form of knowledge: although it does not identify a single distribution, it constrains the range of plausible probabilistic models.

A well-known example of this idea is the exchangeability assumption used in de Finetti's theorem. A sequence of random variables $X_1, X_2, \dots$ is said to be exchangeable if the ordering of the variables carries no information. That is, the joint probability distribution remains invariant under permutations of the variables. This assumption imposes a constraint, effectively introducing a set of plausible distributions. Thus, exchangeability can be seen as a specific form of partial knowledge.

In this work, we adopt the view that a set of probability distributions can represent a meaningful state of *uncertainty* or *partial knowledge*, and that such set-valued uncertainty may itself be used as the basis for inference.

More generally, this partial knowledge may possess a hierarchical structure. In the simplest case, uncertainty is represented by a single set of candidate distributions. In richer settings, however, this uncertainty may be organized into nested collections, where elements of a set may themselves be sets of distributions. Such nesting can continue recursively, yielding a hierarchical structure.

Crucially, the choice of hierarchical organization is itself informative: grouping distributions into subsets expresses that certain alternatives are regarded as belonging to a common epistemic category and should therefore be aggregated before being compared with alternatives in other groups. Therefore, the hierarchy encodes information beyond the collection of leaf distributions alone.

This organization may be interpreted as introducing multiple layers of latent structural decisions, where higher-level choices restrict the admissible lower-level alternatives.

Within a Bayesian framework, inference ultimately requires a single probability measure. Hence, to use hierarchical partial knowledge in Bayesian inference, we require a principled method for transforming a nested set-valued uncertainty into a single probability distribution while preserving the informational structure encoded by the hierarchy.

From a minimax perspective, when uncertainty is represented by a set

of candidate distributions, a natural representative distribution is one that minimizes the worst-case divergence to the members of the set. Such a distribution may be viewed as the most conservative or robust single-distribution summary of the available knowledge.

Notation 1 (Kullback–Leibler Divergence). For probability distributions P and Q , $\mathbb{D}_{\text{KL}}[Q(X) \parallel P(X)]$ denotes the Kullback–Leibler divergence between P and Q over a random variable X . For a discrete random variable X with finite support we have

$$\mathbb{D}_{\text{KL}}[Q(X) \parallel P(X)] = \sum_x Q(x) \ln \frac{Q(x)}{P(x)}.$$

When the underlying random variable is clear from context, we omit the argument and write $\mathbb{D}_{\text{KL}}[Q \parallel P]$.

We assume that the KL divergence is defined for every pair of probability distributions P and Q , taking the value $+\infty$ whenever the support of Q is not contained in the support of P .

Definition 1. Let X be a random variable and $\mathcal{Q}$ be a set of probability distributions over X . We call a distribution C a *central distribution of $\mathcal{Q}$ for X* if, for every probability distribution P , we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(X) \parallel C(X)] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(X) \parallel P(X)],$$

By convention, any probability distribution is regarded as its own central distribution.

Under this definition, the central distribution of $\mathcal{Q}$ is not necessarily an element of $\mathcal{Q}$. That is, the distribution that best represents the set in the specified sense may lie outside the set it summarizes.

Sometimes partial knowledge is represented by a hierarchical structure of sets of distributions. Our current definition of the central distribution does not account for this case. The following recursive definitions extend the concept to handle such situations. While not directly about feature selection, this extension plays a key role in enabling the connection between central distributions and feature selection, a connection that will become clearer later.

Definition 2 (Hierarchy of Distributions). A *hierarchy of distributions* over X is defined recursively as follows:

- Any probability distribution over X is a hierarchy of distributions over X .

- Any set whose elements are hierarchies of distributions over X is itself a hierarchy of distributions over X .

Intuitively, a central distribution acts as a representative or summary of a set of distributions. While it may not belong to the set itself, it reflects the overall properties of the set in a way that is meaningful for inference. It also provides a way to evaluate probability distributions under partial knowledge, motivating the definition of a corresponding central divergence.

The concept of a central distribution is not intended to yield a unique object; rather, all distributions satisfying the definition can be regarded as equally good representatives. However, when multiple central distributions exist, our constructions should not depend on an arbitrary choice among them. To avoid such dependence, we define the central divergence using the best divergence attained over the set of all central distributions.

Definition 3 (Central Divergence). For any distribution P , we define the *central divergence* of P from a hierarchy of distributions $\mathcal{H}$, with respect to a random variable X , as follows:

$$\mathbb{D}_C[\mathcal{H}(X) \parallel P(X)] = \inf_C \mathbb{D}_{\text{KL}}[C(X) \parallel P(X)],$$

where C denotes a central distribution of $\mathcal{H}$ for X .

So far, this definition applies only when the hierarchy is a single set of distributions. The following definition extends the construction recursively to general hierarchical uncertainty structures.

Definition 4 (Central Distribution). Let X be a random variable, and let $\mathcal{H}^*$ be a hierarchy of distributions over X . We call a distribution C a *central distribution of $\mathcal{H}^*$ for X* if, for every probability distribution P , we have:

$$\sup_{\mathcal{H} \in \mathcal{H}^*} \mathbb{D}_C[\mathcal{H}(X) \parallel C(X)] \leq \sup_{\mathcal{H} \in \mathcal{H}^*} \mathbb{D}_C[\mathcal{H}(X) \parallel P(X)].$$

Definition 4 generalizes Definition 1, which is recovered when the hierarchy is a set of probability distributions.

For later use, we introduce a compact notation for the minimax objective appearing in Definition 1.

Notation 2. For a set of probability distributions $\mathcal{Q}$ over X , we define the functional

$$\mathbb{S}_{\mathcal{Q}}[P(X)] := \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(X) \parallel P(X)],$$

where P is an arbitrary probability distribution over X . When $\mathcal{Q}$ and X are clear from the context, we may simply write $\mathbb{S}[P]$.

Notice that this notation allows Definition 1 to be written more compactly as

$$C \in \arg \min_P \mathbb{S}[P].$$

A simple example illustrates that hierarchical organization can affect the resulting representative distribution even when the underlying admissible sets remain unchanged.

Example 2.1. Let X be a random variable with three possible outcomes x_1, x_2, x_3 . For each i , let S_i denote the set of all probability distributions over X that assign zero probability to x_i .

By symmetry, the central distribution of each set S_i is the uniform distribution over its support:

$$C_{S_1} = (0, 0.5, 0.5), \quad C_{S_2} = (0.5, 0, 0.5), \quad C_{S_3} = (0.5, 0.5, 0).$$

If the three sets S_1, S_2, S_3 are treated symmetrically as elements of a single set, then their central distribution is

$$C_{\{S_1, S_2, S_3\}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

Now consider instead a hierarchical organization in which S_1 and S_2 are grouped together, while S_3 forms its own separate branch.

By numerically solving the minimax problem in Definition 4, we obtain

$$C_{\{S_1, S_2\}} = (0.25, 0.25, 0.5).$$

Applying the same procedure to the two distributions $C_{\{S_1, S_2\}}$ and C_{S_3} yields

$$C_{\{\{S_1, S_2\}, S_3\}} = (0.4, 0.4, 0.2).$$

Although the underlying admissible sets are identical in both constructions, the resulting representative distribution differs.

This demonstrates that hierarchical structure carries information beyond the collection of leaf distributions alone. Consequently, flattening a hierarchy into a single unstructured set may discard meaningful epistemic structure.

In many learning problems, uncertainty concerns not only the parameters of a predictive model but also its structural form. For example, when the relevance of potential features is unknown, each candidate feature subset determines a distinct set of plausible models in which the target variable depends only on that feature subset.

Feature selection may therefore be viewed as uncertainty over a collection of distributions. The hierarchical central distribution framework provides a principled mechanism for representing and aggregating such uncertainty. By organizing candidate models hierarchically, one can encode preferences over structural hypotheses, such as favoring sparse subsets, grouping related attributes, or privileging particular interactions.

Candidate feature subsets typically induce overlapping families of admissible distributions. Distributions representing simpler dependency structure may belong simultaneously to many such families, whereas more complex dependency patterns belong to fewer. As a result, hierarchical aggregation through central distributions can induce a preference toward structurally simpler models by naturally assigning more weight to distributions compatible with multiple structural hypotheses.

2.1 Minimax Characterization

With the concept of hierarchical central distributions in place, we now develop the theoretical machinery needed to characterize and compute them. Throughout this subsection, X denotes a discrete random variable with finite support and $\mathcal{Q}$ denotes a finite set of probability distributions over X .

As we shall see, central distributions arise naturally as solutions to a convex–concave game. By formulating the appropriate objective function, we obtain a saddle-point characterization that enables the use of standard techniques from convex optimization.

Definition 5 (Saddle Point). Consider a function $f : \mathcal{W} \times \mathcal{Z} \rightarrow \mathbb{R}$. We refer to a pair (w^*, z^*) as a *saddle point* for f if we have

$$f(w^*, z) \leq f(w^*, z^*) \leq f(w, z^*),$$

for all $Q \in \mathcal{W}$ and $Q \in \mathcal{Z}$.

Lemma 1. *Define:*

$$f(P, R) := \sum_i \mathbb{D}_{\text{KL}} [Q_i(X) \parallel P(X)] R(i),$$

where P and R are probability distributions, $Q_i \in \mathcal{Q}$ and the index i ranges over all elements of $\mathcal{Q}$.

There exists a saddle point for f .

Proof. The Kullback–Leibler divergence is convex and lower-semicontinuous in its second argument. Therefore, for any fixed probability distribution R ,

the function $f(\cdot, R)$ is a finite convex combination of lower-semicontinuous convex functions and is therefore itself convex and lower-semicontinuous in P .

For fixed P , the function $f(P, \cdot)$ is affine in R , and hence concave and continuous.

The set of probability distributions over a discrete random variable with finite support is a compact convex simplex. Therefore, the assumptions of Sion's minimax theorem are satisfied [6], and

$$\min_P \sup_R f(P, R) = \sup_R \min_P f(P, R).$$

Since $f(P, \cdot)$ is continuous and the feasible set for R is compact, the supremum is attained for every P . Hence the minimax solution exists and f admits a saddle point. $\square$

Lemma 2. *Let f be the function defined in Lemma 1. If (P^*, R^*) is a saddle point for f , then P^* is a central distribution of $\mathcal{Q}$ for X .*

Proof. Let P be an arbitrary probability distribution over X , and (P^*, R^*) be a saddle point for f . As P is a probability distribution, $f(P, R^*)$ is defined, and by Definition 5

$$f(P^*, R^*) \leq f(P, R^*). \quad (1)$$

Since R^* is a probability distribution, the definition of f implies:

$$f(P, R^*) \leq \mathbb{S}[P].$$

On the other hand, by the definition of a saddle point:

$$f(P^*, R^*) = \max_R f(P^*, R) = \mathbb{S}[P^*].$$

Hence, Equation 1 implies

$$\mathbb{S}[P^*] \leq \mathbb{S}[P].$$

This holds for any arbitrary P . Therefore, by Definition 1, P^* is a central distribution of $\mathcal{Q}$ for X . $\square$

Lemma 3. *Let f be the function defined in Lemma 1. The point (P^*, R^*) is a saddle point for f , if and only if there is a constant λ , and a function μ ,*

such that for every x and i the following conditions hold:

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} + \mu(i) = \lambda, \quad (2)$$

$$P^*(x) = \sum_i Q_i(x) R^*(i) \quad (3)$$

$$\sum_i R^*(i) = 1, \quad (4)$$

$$R^*(i) \geq 0, \quad (5)$$

$$\mu(i) \geq 0, \quad (6)$$

$$\mu(i) R^*(i) = 0. \quad (7)$$

Furthermore, for a fixed probability distribution R , a distribution $\hat{P}$ minimizes $f(\cdot, R)$, if and only if for all x it satisfies

$$\hat{P}(x) = \sum_i Q_i(x) R(i). \quad (8)$$

Proof. We extend f to a function $\hat{f}$ defined as follows:

$$\hat{f}(P, R) := \sum_i R(i) \sum_x Q_i(x) \ln \frac{Q_i(x)}{P(x)},$$

where the domain is extended by relaxing the constraints that P and R be probability distributions. This enables us to formulate the saddle points of f as solutions to a constrained optimization problem defined for $\hat{f}$.

Let (P^*, R^*) be a simultaneous solution to the following two optimization problems:

$$\begin{aligned} & \text{minimize} && \hat{f}(P, R) && \text{w.r.t. } P \\ & \text{subject to} && \sum_x P(x) = 1, \\ & && && \\ & \text{maximize} && \hat{f}(P, R) && \text{w.r.t. } R \\ & \text{subject to} && R(i) \geq 0, && \text{for all } i \\ & && \sum_i R(i) = 1 \end{aligned} \quad (9)$$

If (P^*, R^*) exists, it will be a saddle point for f .

Both optimization problems are convex and differentiable; see [3]. To establish this, consider that all the constraint functions are affine. Additionally, both $\hat{f}(P, \cdot)$ and $\hat{f}(\cdot, R)$ are differentiable on their domains. For any fixed P , $\hat{f}(P, \cdot)$ is an affine function and is concave. For a fixed solution to the second optimization problem denoted by $\hat{R}$, we can write

$$\hat{f}(P, \hat{R}) = c - \sum_{i,x} \hat{R}(i) Q_i(x) \ln P(x),$$

where c is not a function of P . Since $\hat{R}$ is a solution to the second optimization problem, it satisfies the corresponding constraints and is therefore a valid probability distribution. Hence, for all i, x we have $\hat{R}(i) Q_i(x) \geq 0$, and $\hat{f}(\cdot, \hat{R})$ is the negative of a linear combination of concave $\ln P(x)$ functions with nonnegative coefficients. This function is therefore convex.

Thus, both optimization problems are convex and differentiable, with affine equality and inequality constraints. In this setting, Slater's condition reduces to feasibility [3]. The feasible set for the first problem is the probability simplex, while the feasible set for the second problem is the strictly positive probability simplex. Both sets are nonempty. Therefore, Slater's condition holds, and the KKT conditions are necessary and sufficient for optimality. In other words, if there exists a point (P^*, R^*) that satisfies the KKT conditions for both problems, it will be a saddle point for f , and vice versa.

The KKT conditions are as follows. There exist constants λ, η and a function μ , such that for every x and i :

$$\sum_i R^*(i) Q_i(x) = \eta P^*(x) \tag{10}$$

$$\sum_x P^*(x) = 1, \tag{11}$$

$$\begin{aligned} \sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} + \mu(i) &= \lambda, \\ \sum_i R^*(i) &= 1, \\ R^*(i) &\geq 0, \\ \mu(i) &\geq 0, \\ \mu(i) R^*(i) &= 0. \end{aligned} \tag{12}$$

By summing Equation 10 over x and using Equation 12, we obtain $\eta = 1$. Substituting this into Equation 10 yields Equation 3, which in turn implies Equation 11.

To prove the second part of the lemma, we note that Equations 10 and 11 are the KKT conditions for the minimization problem in Equation 9. As shown above, the KKT conditions are necessary and sufficient for optimality, and when R is a probability distribution, these conditions simplify to

$$\hat{P}(x) = \sum_i Q_i(x) R(i). \quad \square$$

Lemma 4. *Let P be a convex combination of the elements of $\mathcal{Q}$ such that*

$$P(x) = \sum_i Q_i(x) R(i),$$

where R is a probability distribution over the index set of $\mathcal{Q}$. Let f be the function defined in Lemma 1. Define the function g as:

$$g(P) := f(P, R).$$

If (P^, R^*) is a saddle point for f , then P^* is a maximizer of g .*

Proof. Let (P^*, R^*) be a saddle point for f . By Lemma 3, P^* satisfies Equation 3 and R^* satisfies Equations 4 and 5. Therefore,

$$g(P^*) = f(P^*, R^*).$$

Now, let P be an arbitrary convex combination of the elements of $\mathcal{Q}$, and let R denote the corresponding weights. From the second part of Lemma 3, it follows that $g(P)$ indeed corresponds to the minimum of $f(\cdot, R)$, which implies

$$g(P) \leq f(P^*, R).$$

Since (P^*, R^*) is a saddle point, by Definition 5,

$$f(P^*, R) \leq f(P^*, R^*).$$

Combining the two inequalities, we get:

$$g(P) \leq g(P^*).$$

This shows that $g(P^*)$ is the maximum of g , as desired. $\square$

Lemma 5. *Let $\mathcal{R} \subseteq \mathcal{Q}$, let P be a convex combination of the elements of $\mathcal{R}$, and let P' be any probability distribution. Then*

$$\inf_{Q \in \mathcal{R}} \mathbb{D}_{\text{KL}}[Q \| P'] \leq \mathbb{D}_{\text{KL}}[P \| P'] + g(P) \leq \sup_{Q \in \mathcal{R}} \mathbb{D}_{\text{KL}}[Q \| P'],$$

where g is the function defined in Lemma 4.

Proof. Let R denote the weights associated with the convex representation of P . For any probability distribution P'

$$\begin{aligned}\mathbb{D}_{\text{KL}}[P \parallel P'] &= \sum_i R(i) (\mathbb{D}_{\text{KL}}[Q_i \parallel P'] - \mathbb{D}_{\text{KL}}[Q_i \parallel P]) \\ &= \sum_i \mathbb{D}_{\text{KL}}[Q_i \parallel P'] R(i) - g(P),\end{aligned}$$

where i ranges over the elements of $\mathcal{R}$. Because R is a probability distribution, we have

$$\inf_{Q \in \mathcal{R}} \mathbb{D}_{\text{KL}}[Q \parallel P'] \leq \sum_i \mathbb{D}_{\text{KL}}[Q_i \parallel P'] R(i) \leq \sup_{Q \in \mathcal{R}} \mathbb{D}_{\text{KL}}[Q \parallel P'],$$

which proves the lemma. $\square$

The following theorem establishes the existence and uniqueness of the central distribution and provides a characterization of its optimality in the minimax setting. This characterization motivates the term *central distribution*: like the center of a circle, the central distribution is equidistant from the boundary.

Theorem 1 (Centrality). *Let X be a discrete random variable with finite support, and let $\mathcal{Q}$ be a finite set of probability distributions over X . Then $\mathcal{Q}$ has a unique central distribution for X .*

Moreover, C is the central distribution of $\mathcal{Q}$ for X if and only if it can be expressed as a convex combination of the elements of $\mathcal{Q}$; that is,

$$C(x) = \sum_i Q_i(x) R(i),$$

where R is a probability distribution over the index set of $\mathcal{Q}$, and there exists a constant λ such that

$$\mathbb{D}_{\text{KL}}[Q_i(X) \parallel C(X)] = \lambda$$

whenever $R(i) > 0$, and

$$\mathbb{D}_{\text{KL}}[Q_i(X) \parallel C(X)] \leq \lambda$$

whenever $R(i) = 0$.

Proof. By Lemma 1 the function f has a saddle point. Let (C, R) denote this saddle point. By Lemma 2, C is a central distribution of $\mathcal{Q}$ for X .

The saddle point (C, R) satisfies the conditions of Lemma 3. Equations 3, 4 and 5 imply that C is a convex combination of the elements of $\mathcal{Q}$. If $R(i) > 0$, then Equation 7 implies that $\mu(i) = 0$. Substituting into Equation 2 yields

$$\mathbb{D}_{\text{KL}}[Q_i \parallel C] = \lambda.$$

If $R(i) = 0$, then Equation 6 implies that $\mu(i) \geq 0$. Combining this with Equation 2 gives

$$\mathbb{D}_{\text{KL}}[Q_i \parallel C] \leq \lambda.$$

Since C is a convex combination of the elements of $\mathcal{Q}$, and $g(C) = \lambda$, by Lemma 5, for any probability distribution C' , we have

$$\mathbb{D}_{\text{KL}}[C \parallel C'] \leq \mathbb{S}[C'] - \lambda.$$

Now suppose that C' is also a central distribution of $\mathcal{Q}$ for X . Because C and C' are both central distributions,

$$\mathbb{S}[C'] = \mathbb{S}[C] = \lambda.$$

Therefore, by non-negativity of the KL divergence, we have

$$\mathbb{D}_{\text{KL}}[C \parallel C'] = 0,$$

which implies C and C' are equal probability distributions.

Thus, we have proved that $\mathcal{Q}$ has a unique central distribution satisfying the conditions of the theorem. We now prove the “if” direction. Assume that the conditions of the theorem hold for C . Since C is a convex combination of the elements of $\mathcal{Q}$ it is a probability distribution over X . By defining

$$\mu(i) = \lambda - \mathbb{D}_{\text{KL}}[Q_i \parallel C],$$

we have $\mu(i) \geq 0$ and $\mu(i)R(i) = 0$ for every index i . We can verify that C and R satisfy the conditions of Lemma 3 and therefore (C, R) is a saddle point for f . By Lemma 2, C is a central distribution of $\mathcal{Q}$ for X . $\square$

Corollary 1.1. *Let C be the central distribution of $\mathcal{Q}$ for X . For any probability distribution P , we have*

$$\mathbb{D}_{\text{C}}[\mathcal{Q}(X) \parallel P(X)] \leq \mathbb{S}_{\mathcal{Q}}[P(X)] - \mathbb{S}_{\mathcal{Q}}[C(X)].$$

Proof. By Theorem 1, C is unique, and therefore

$$\mathbb{D}_{\text{C}}[\mathcal{Q} \parallel P] = \mathbb{D}_{\text{KL}}[C \parallel P].$$

Moreover, C is a convex combination of the elements of $\mathcal{Q}$, and $g(C) = \mathbb{S}[C]$. The result then follows immediately from Lemma 5. $\square$

Theorem 2. *The central distribution of $\mathcal{Q}$ for X is a maximizer of the function g defined in Lemma 4.*

Proof. By Lemmas 1 and 4, there exists a saddle point (C, R) for f , where C is a maximizer of the function g . By Lemma 2, C is a central distribution of $\mathcal{Q}$ for X , and Theorem 1 implies that C is unique. $\square$

Corollary 2.1. *Let C be the central distribution of $\mathcal{Q}$ for X , and let P be a convex combination of the elements of $\mathcal{R} \subseteq \mathcal{Q}$. Then there exists $Q_k \in \mathcal{R}$ such that*

$$\mathbb{S}_{\mathcal{Q}}[C(X)] \geq \mathbb{D}_{\text{KL}}[Q_k(X) \parallel P(X)].$$

Proof. We proceed by contradiction. Suppose that for every $Q \in \mathcal{R}$, we have $\mathbb{D}_{\text{KL}}[Q \parallel P] > \mathbb{S}[C]$. That implies

$$g(P) > \mathbb{S}[C],$$

where g is the function defined in Lemma 4. Note that $g(P)$ is defined, because P is a convex combination of the elements of $\mathcal{Q}$.

Theorem 1 implies that $g(C) = \mathbb{S}[C]$. However, by Theorem 2, the central distribution C maximizes g , which contradicts the inequality $g(P) > g(C)$. $\square$

2.2 Parametric Families and Central Priors

The results of the previous subsection were established under the assumption that $\mathcal{Q}$ was finite. Many of the underlying constructions, however, can be extended to parametric families in which $\mathcal{Q}$ is indexed by a finite-dimensional parameter vector Θ :

$$\mathcal{Q} = \{Q_{\theta} \mid \theta \in \text{val}(\Theta)\}.$$

In this setting, Θ may be regarded as a random variable, with each Q_{θ} interpreted as a conditional distribution $Q(x \mid \theta)$. The weight function $R(i)$ appearing in the finite case is then replaced by a prior distribution over Θ .

A rigorous development of this extension requires additional measure-theoretic and regularity assumptions and is beyond the scope of the present paper.

Definition 6 (Central Prior). Let X be a random variable, and let $\mathcal{Q}$ be a set of probability distributions over X , indexed by a parameter Θ . A nonnegative measure c on Θ is called a central prior of $\mathcal{Q}$ for X if,

$$C(x) = \int Q(x \mid \theta) c(\theta) d\theta$$

is a central distribution of $\mathcal{Q}$ for X .

Definition 7 (Interior Distribution). Let $\mathcal{Q}$ be a set of probability distributions over a random variable X . A distribution $R \in \mathcal{Q}$ is called an *interior distribution* of $\mathcal{Q}$ if, the sets $\mathcal{Q}$ and $\mathcal{Q} \setminus \{R\}$ have the same central distributions for X .

Proposition 1. Let X be a discrete random variable with finite support, and let $\mathcal{Q}$ be a finite set of probability distributions over X . Let C be the central distribution of $\mathcal{Q}$ for X . If for some $Q_k \in \mathcal{Q}$

$$\mathbb{D}_{\text{KL}}[Q_k \| C] < \mathbb{S}_{\mathcal{Q}}[C], \quad (13)$$

then Q_k is an interior distribution of $\mathcal{Q}$.

Proof. By Theorem 1, C is a convex combination of elements of $\mathcal{Q}$, and Equation 13 implies that $R(k) = 0$. Hence, C is also a convex combination of elements of $\mathcal{Q} \setminus \{Q_k\}$ and the conditions of Theorem 1 remain satisfied for C and $\mathcal{Q} \setminus \{Q_k\}$. Thus, C is the unique central distribution of $\mathcal{Q} \setminus \{Q_k\}$ and $\mathcal{Q}$ for X . $\square$

Theorem 3. Let X be a random variable and let $\mathcal{Q}$ be a set of probability distributions over X . If $R \in \mathcal{Q}$ can be expressed as a convex combination of the elements of $\mathcal{Q} \setminus \{R\}$, then R is an interior distribution of $\mathcal{Q}$.

Proof. Let $R = \sum_i \alpha_i Q_i$, where $Q_i \in \mathcal{Q} \setminus \{R\}$, $\alpha_i \geq 0$, and $\sum_i \alpha_i = 1$.

Fix an arbitrary probability distribution P . By the convexity of the KL divergence in its first argument, we have

$$\mathbb{D}_{\text{KL}}[R \| P] \leq \sum_i \alpha_i \mathbb{D}_{\text{KL}}[Q_i \| P].$$

Note that the right-hand side is a weighted average of the KL divergences. So, there must be at least one index k such that

$$\mathbb{D}_{\text{KL}}[R \| P] \leq \mathbb{D}_{\text{KL}}[Q_k \| P].$$

Since $Q_k \in \mathcal{Q} \setminus \{R\}$, it follows that

$$\mathbb{S}_{\mathcal{Q}}[P] = \mathbb{S}_{\mathcal{Q} \setminus \{R\}}[P]$$

for every probability distribution P . Therefore, $\mathcal{Q}$ and $\mathcal{Q} \setminus \{R\}$ have the same central distributions. $\square$

One important consequence of this theorem is that, in order to identify a central distribution within a parametric model, it suffices to restrict attention to the set of likelihood distributions induced by different values of the model parameters.

Theorem 4. *Let X and T be two random variables, and let $\mathcal{Q}$ be a set of distributions over the pair (X, T) . Assume that for every $Q \in \mathcal{Q}$, we have*

$$Q(X, T) = R(X | T)Q(T),$$

where R is a fixed conditional distribution. If C is a central distribution of $\mathcal{Q}$ for T , then $R(X | T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T) .

Proof. Let C be a central distribution of $\mathcal{Q}$ for T . For any arbitrary probability distribution P , the definition of a central distribution implies

$$\mathbb{S} [R(X | T)C(T)] \leq \mathbb{S} [R(X | T)P(T)]. \quad (14)$$

Since the KL divergence is nonnegative, for any probability distribution P and every $Q \in \mathcal{Q}$ we have

$$\sum_t \mathbb{D}_{\text{KL}} [R(X | t) \| P(X | t)] Q(t) \geq 0.$$

That implies

$$\mathbb{D}_{\text{KL}} [Q(X, T) \| R(X | T)P(T)] \leq \mathbb{D}_{\text{KL}} [Q(X, T) \| P(X, T)].$$

Taking the supremum over $Q \in \mathcal{Q}$ preserves the inequality, since it holds pointwise for every $Q \in \mathcal{Q}$:

$$\mathbb{S} [R(X | T)P(T)] \leq \mathbb{S} [P(X, T)].$$

Combining this inequality with Equation 14, we conclude that $R(X | T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T) . $\square$

A particularly important special case arises when T is a deterministic function of X , say $T = f(X)$. In this case

$$P(X = x) = P(X = x, T = f(x)),$$

and a central distribution for X can be constructed directly from a central distribution for T using Theorem 4. Therefore, the primary application of this theorem is to derive central distributions for variables of interest from central distributions of sufficient statistics.

The following theorem provides a sufficient fixed point characterization of central priors and highlights a formal connection between the central distribution framework and Bernardo's reference prior methodology [2].

Theorem 5 (Fixed Point). *Let X be a discrete random variable, and let $\mathcal{Q}$ be a set of probability distributions over X , indexed by a finite-dimensional vector Θ . Let c be a (possibly improper) prior distribution for Θ such that*

$$c(\theta) \propto \exp \left\{ \sum_x Q(x | \theta) \ln c(\theta | x) \right\}. \quad (15)$$

Then, c is a central prior of $\mathcal{Q}$ for X .

Proof. Since $Q_\theta \in \mathcal{Q}$ is a probability distribution, for every θ there exists x_0 such that $Q_\theta(x_0) > 0$. Equation 15 therefore implies $c(\theta | x_0) > 0$, and consequently $c(\theta) > 0$.

Using Equation 15 and $c(\theta) > 0$, we obtain

$$\begin{aligned} c(\theta) &= \exp \left\{ -\lambda + \sum_x Q(x | \theta) \ln \frac{Q(x | \theta)c(\theta)}{C(x)} \right\} \\ &= \exp \{ -\lambda + \mathbb{D}_{\text{KL}} [Q_\theta(X) \| C(X)] + \ln c(\theta) \} \\ &= c(\theta) \exp \{ -\lambda + \mathbb{D}_{\text{KL}} [Q_\theta(X) \| C(X)] \}, \end{aligned}$$

and it follows that

$$\mathbb{D}_{\text{KL}} [Q_\theta(X) \| C(X)] = \lambda.$$

Since this holds for every θ , and λ is constant, a parametric extension of Theorem 1 implies that C , the marginal distribution induced by c , is a central distribution of $\mathcal{Q}$ for X . Therefore, by Definition 6, c is a central prior of $\mathcal{Q}$ for X . $\square$

This theorem is proved using Theorem 1. However, Theorem 2 suggests an alternative route to establishing the connection with reference priors. Since reference priors are defined by maximizing missing information, the optimization characterization provided by Theorem 2 also offers another perspective on this relationship.

2.3 Asymptotic Central Distributions

Many probabilistic models are naturally formulated in terms of a sequence of random variables whose length is not fixed in advance. For example, when a coin is tossed repeatedly, there is no natural upper bound on the number of tosses, and any finite sample can be viewed as an initial segment of an infinite sequence.

Motivated by such examples, let S_n denote the first n variables of an infinite sequence of random variables:

$$S_n = (X_1, X_2, \dots, X_n).$$

The collection $\{S_n\}_{n=1}^{\infty}$ therefore defines a second sequence of random variables.

Given n observations, we may consider a set of admissible distributions over S_n and then construct a central distribution of that set for S_n . However, the resulting central distribution generally depends on the sequence length n .

We believe that the probability measure used to represent partial knowledge about an environment should depend only on the available information about that environment, and not on arbitrary aspects of the experimental setup, such as the number of variables under consideration. This motivates the search for a single probability measure that remains representative across all sequence lengths.

Importantly, a distribution defined over a longer sequence induces marginal distributions over all shorter initial segments. This suggests that, rather than constructing a separate central distribution for each S_n , we should seek a single probability measure over the infinite sequence, from which the distributions of all finite initial segments can be obtained as marginals.

This motivates the construction of a central distribution that is well defined for S_{∞} .

Definition 8 (Asymptotic Central Distribution). Let $\{S_n\}_{n=1}^{\infty}$ be a sequence of random variables and $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n . Let C be also a probability measure that defines a distribution over every S_n . C is called a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$ if,

$$\lim_{n \rightarrow \infty} \mathbb{D}_C [\mathcal{Q}(S_n) \parallel C(S_n)] = 0.$$

Intuitively, this definition states that as $n \rightarrow \infty$ the distribution that C induces over S_n becomes increasingly similar to a central distribution for S_n . In other words, C is a central distribution for the infinite sequence S_{∞} .

Proposition 2 (Asymptotic Uniqueness). *Suppose there exists M such that for every $n \geq M$, $\mathcal{Q}$ has a unique central distribution C_n for S_n . Then any two central distributions C and C' of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$, are asymptotically equivalent in total variation:*

$$\lim_{n \rightarrow \infty} \|C(S_n) - C'(S_n)\|_{\text{TV}} = 0.$$

Here, $\|\cdot\|_{\text{TV}}$ denotes total variation distance.

Proof. Let C and C' be two central distributions of $\mathcal{Q}$ for $\{S_n\}_{n=1}^\infty$. Since, for $n \geq M$, C_n is the unique central distribution of $\mathcal{Q}$ for S_n , we have

$$\lim_{n \rightarrow \infty} \mathbb{D}_{\text{KL}}[C_n(S_n) \parallel C(S_n)] = \lim_{n \rightarrow \infty} \mathbb{D}_{\text{KL}}[C_n(S_n) \parallel C'(S_n)] = 0.$$

By Pinsker's inequality and the squeeze theorem,

$$\lim_{n \rightarrow \infty} \|C_n(S_n) - C(S_n)\|_{\text{TV}} = \lim_{n \rightarrow \infty} \|C_n(S_n) - C'(S_n)\|_{\text{TV}} = 0.$$

Using the triangle inequality,

$$\|C(S_n) - C'(S_n)\|_{\text{TV}} \leq \|C_n(S_n) - C(S_n)\|_{\text{TV}} + \|C_n(S_n) - C'(S_n)\|_{\text{TV}},$$

the squeeze theorem yields

$$\lim_{n \rightarrow \infty} \|C(S_n) - C'(S_n)\|_{\text{TV}} = 0. \quad \square$$

Our next goal is to extend Theorem 1 so that it also applies in this asymptotic setting.

Theorem 6 (Asymptotic Centrality). *Let $\{S_n\}_{n=1}^\infty$ be a sequence of discrete random variables with finite support and let $\mathcal{Q}$ be a finite set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n . Suppose that C is a convex combination of the elements of $\mathcal{Q}$. If there exists a sequence of constants $\{\lambda_n\}_{n=1}^\infty$ such that*

$$\lim_{n \rightarrow \infty} \sup_{Q \in \mathcal{Q}} |\mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] - \lambda_n| = 0,$$

then C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^\infty$, and is unique up to asymptotic total variation equivalence.

Proof. For every n , let C_n be the unique central distribution of $\mathcal{Q}$ for S_n , and let μ_n denote the corresponding constant from Theorem 1. Since C is a convex combination of elements of $\mathcal{Q}$, it defines a distribution over S_n , and Corollary 2.1 implies

$$\mu_n \geq \inf_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)],$$

By the definition of a central distribution,

$$\mu_n \leq \mathbb{S}[C(S_n)].$$

Moreover, we have

$$\left| \inf_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(S_n) \parallel C(S_n)] - \lambda_n \right| \leq \sup_{Q \in \mathcal{Q}} |\mathbb{D}_{\text{KL}} [Q(S_n) \parallel C(S_n)] - \lambda_n|,$$

and

$$|\mathbb{S} [C(S_n)] - \lambda_n| \leq \sup_{Q \in \mathcal{Q}} |\mathbb{D}_{\text{KL}} [Q(S_n) \parallel C(S_n)] - \lambda_n|.$$

Hence, both $\{\inf_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(S_n) \parallel C(S_n)] - \lambda_n\}_{n=1}^{\infty}$ and $\{\mathbb{S} [C(S_n)] - \lambda_n\}_{n=1}^{\infty}$ converge to 0. By the squeeze theorem, the sequence $\{\mu_n - \lambda_n\}_{n=1}^{\infty}$ also converges to 0. We may write

$$\begin{aligned} \lim_{n \rightarrow \infty} (\mathbb{S} [C(S_n)] - \mu_n) &= \lim_{n \rightarrow \infty} (\mathbb{S} [C(S_n)] - \lambda_n) - \lim_{n \rightarrow \infty} (\mu_n - \lambda_n) \\ &= 0. \end{aligned}$$

By Corollary 1.1, we have

$$\mathbb{D}_{\text{C}} [\mathcal{Q}(S_n) \parallel C(S_n)] \leq \mathbb{S} [C(S_n)] - \mu_n,$$

and the squeeze theorem implies

$$\lim_{n \rightarrow \infty} \mathbb{D}_{\text{C}} [\mathcal{Q}(S_n) \parallel C(S_n)] = 0.$$

Thus, by Definition 8, C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$. Since for every n , C_n is the unique central distribution of $\mathcal{Q}$ for S_n , Proposition 2 implies that the central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^{\infty}$ is unique up to asymptotic total variation equivalence. $\square$

We present the following conjecture as a heuristic framework for understanding the asymptotic form of central priors. The argument is inspired by a similar asymptotic approach in [2].

Conjecture. Let $\{T_n\}_{n=1}^{\infty}$ be a sequence of random variables, and let $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each T_n . Assume that $\mathcal{Q}$ is indexed by a finite-dimensional vector Θ , and $\{T_n\}_{n=1}^{\infty}$ is a consistent sequence of estimators for Θ . Define

$$c(\theta) \propto \lim_{n \rightarrow \infty} \frac{P(\theta \mid T_n = t_n)|_{t_n=\theta}}{\mu_n}, \quad (16)$$

where $\{\mu_n\}_{n=1}^{\infty}$ is a sequence of positive constants and $P(\theta \mid T_n)$ denotes the posterior distribution obtained from an arbitrary smooth positive prior. If this limit exists and the convergence is uniform over θ , under the usual regularity conditions, $c(\theta)$ is a central prior of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$.

Let k denote the normalizing constant converting proportionality into equality. Since the convergence is uniform and the logarithm is a continuous function, Equation 16 implies

$$\lim_{n \rightarrow \infty} \sup_{\theta} \left| \ln \frac{P(\theta \mid T_n = t_n)|_{t_n=\theta}}{c(\theta)} - \ln \mu_n + \ln k \right| = 0.$$

Because T_n is a consistent estimator of every θ , the sampling distribution $Q(T_n \mid \theta)$ becomes increasingly concentrated around θ as $n \rightarrow \infty$. Therefore, we may write

$$\lim_{n \rightarrow \infty} \sup_{\theta} \left| \sum_{t_n} Q(t_n \mid \theta) \ln \frac{P(\theta \mid t_n)}{c(\theta)} - \ln \mu_n + \ln k \right| = 0.$$

Furthermore, under the usual regularity conditions, the posterior distribution becomes asymptotically insensitive to the choice of smooth positive prior. Motivated by this, we heuristically replace $P(\theta \mid T_n)$ with $c(\theta \mid T_n)$, obtaining

$$\lim_{n \rightarrow \infty} \sup_{\theta} \left| \sum_{t_n} Q(t_n \mid \theta) \ln \frac{Q(t_n \mid \theta)}{C(t_n)} - \ln \mu_n + \ln k \right| = 0,$$

where

$$C(t_n) = \int Q(t_n \mid \theta) c(\theta) d\theta.$$

Letting $\lambda_n = \ln \mu_n - \ln k$, we observe that

$$\lim_{n \rightarrow \infty} \sup_{\theta} |\mathbb{D}_{\text{KL}}[Q(T_n \mid \theta) \parallel C(T_n)] - \lambda_n| = 0.$$

By a parametric extension of Theorem 6, it can be shown that C is a central distribution of $\mathcal{Q}$ for $\{T_n\}_{n=1}^{\infty}$, and therefore $c(\theta)$ is a central prior for the estimator sequence $\{T_n\}_{n=1}^{\infty}$.

Example 2.2 (Binomial Experiment). Consider a binomial experiment consisting of n independent Bernoulli trials. Let $\mathcal{Q}$ denote the family of probability distributions corresponding to different values of the success probability $\theta \in [0, 1]$, with θ serving as the indexing parameter for $\mathcal{Q}$. Let T_n be the sample proportion of successes in the n trials. Since T_n is a consistent estimator of θ , we can apply Equation 16 to determine a central distribution for the sequence $\{T_n\}_{n=1}^{\infty}$.

Assuming a uniform prior on θ , Bayes' rule gives the posterior distribution

$$P(\theta \mid T_n = \frac{k}{n}) = (n+1) \binom{n}{k} \theta^k (1-\theta)^{n-k},$$

which is a Beta($k + 1, n - k + 1$) distribution. Using Stirling's formula, we derive its asymptotic form:

$$P\left(\theta \mid T_n = \frac{k}{n}\right) \sim \frac{(n+1)n^{(n+\frac{1}{2})}}{\sqrt{2\pi}(n-k)^{(n-k+\frac{1}{2})}k^{(k+\frac{1}{2})}} \theta^k (1-\theta)^{n-k}.$$

Evaluating the asymptotic expression at $\frac{k}{n} = \theta$, we obtain

$$P(\theta \mid T_n = \frac{k}{n})|_{\frac{k}{n}=\theta} \sim \frac{n+1}{\sqrt{2\pi n}} \theta^{-\frac{1}{2}} (1-\theta)^{-\frac{1}{2}}.$$

Therefore, if we define

$$\mu_n = \frac{n+1}{\sqrt{2\pi n}},$$

the following limit exists and the convergence is uniform over $\theta \in (0, 1)$:

$$\lim_{n \rightarrow \infty} \frac{P(\theta \mid T_n = t_n)|_{t_n=\theta}}{\mu_n} = \theta^{-\frac{1}{2}} (1-\theta)^{-\frac{1}{2}}.$$

A central distribution over an estimator sequence cannot, by itself, be used to perform inference at the level of observed outcomes. To do so, we must convert it into a distribution over the primary variables of interest. Theorem 4 provides a simple method for this conversion.

Example 2.3 (Fused Multinomial Model). Let $\{S_n\}_{n=1}^{\infty}$ be a sequence in which each $S_n = (X_1, X_2, \dots, X_n)$ represents n independent trials of an experiment with k possible outcomes, and $\mathcal{Q}$ be a set of probability distributions over S_n indexed by a real vector θ . Let $f: \mathcal{X} \rightarrow \mathcal{Y}$ be a deterministic function that maps each outcome of X_i to a label $Y_i = f(X_i)$, where $\mathcal{Y} = \{1, \dots, m\}$. Assume that for all i and $Q_\theta \in \mathcal{Q}$, if $f(x_i) = f(x'_i)$ we have

$$Q_\theta(x_1, x_2, \dots, x_n) = Q_\theta(x'_1, x'_2, \dots, x'_n).$$

That is, the conditional distribution $Q_\theta(X_i \mid Y_i = y)$ is uniform over the preimage $f^{-1}(y)$.

We consider $\theta = (\theta_1, \dots, \theta_m)$ to be a probability vector over $\mathcal{Y}$, where each θ_y specifies the probability of observing label y . For a sequence S_n , let n_y denote the number of times label $y \in \mathcal{Y}$ appears. Let $|f^{-1}(y)|$ denote the number of outcomes in $\mathcal{X}$ that are mapped to label y by f . Then the probability of observing S_n under $Q_\theta \in \mathcal{Q}$, denoted $Q(S_n \mid \theta)$, is:

$$Q(S_n \mid \theta) = \prod_{y=1}^m \left(\frac{1}{|f^{-1}(y)|} \cdot \theta_y \right)^{n_y}.$$

Consider the estimator sequence $T_n = \frac{1}{n} (n_1, n_2, \dots, n_m)$, which is a consistent estimator of θ . The conditional distribution $Q_\theta(S_n \mid T_n)$ is given by:

$$R(S_n \mid T_n) = \frac{1}{n!} \prod_{y=1}^m \frac{n_y!}{|f^{-1}(y)|^{n_y}}. \quad (17)$$

Note that this distribution is independent of θ . Therefore, a central distribution for S_n can be obtained from a central distribution for T_n .

As $n \rightarrow \infty$, the distribution of T_n converges to a multivariate Gaussian distribution. Using this fact and Equation 16, we can determine a central prior for $\{T_n\}_{n=1}^\infty$:

$$c(\theta) \propto \prod_{y=1}^m \theta_y^{-\frac{1}{2}}.$$

This is a Dirichlet prior and induces a Dirichlet-multinomial distribution for each T_n :

$$C(T_n) = \frac{\Gamma(\frac{m}{2}) \Gamma(n+1)}{\Gamma(n + \frac{m}{2})} \prod_{y=1}^m \frac{\Gamma(n_y + \frac{1}{2})}{\Gamma(\frac{1}{2}) \Gamma(n_y + 1)}.$$

Here, C is a central distribution for the sequence $\{T_n\}_{n=1}^\infty$, viewed as a probability measure on the infinite sequence T_∞ . The notation $C(T_n)$ denotes the marginal distribution of T_n .

By Theorem 4 and Equation 17, we can derive a central distribution for $\{S_n\}_{n=1}^\infty$, which induces the following distribution over S_n :

$$C(S_n) = \frac{\Gamma(\frac{m}{2})}{\Gamma(n + \frac{m}{2})} \prod_{y=1}^m \frac{\Gamma(n_y + \frac{1}{2})}{\Gamma(\frac{1}{2}) |f^{-1}(y)|^{n_y}}.$$

When dealing with layered ignorance and seeking the central distribution of a hierarchy of distributions, we typically encounter situations in which a given level of the hierarchy contains only finitely many alternatives. In such cases, and under appropriate regularity conditions, if there exists a consistent estimator for the index identifying these alternatives, then the central distribution reduces to an equally weighted average of the corresponding representative central distributions.

2.4 Approximate Central Distributions

Suppose we want to approximate a central distribution of a hierarchy of distributions $\mathcal{H}$. To do so, we need a measure that quantifies the quality of an approximation. A natural choice for this purpose is the central divergence introduced in Definition 3. Since central distributions are intended to

serve as representatives of hierarchies of distributions, central divergence can be viewed as a measure of how well a probability distribution represents a hierarchy.

At first glance, computing this quantity appears to require explicit knowledge of the central distributions of $\mathcal{H}$. However, it turns out that the central divergence admits an upper bound that can be estimated without explicitly knowing the central distribution.

In variational settings, it is often more convenient to work with unnormalized measures rather than normalized distributions. Therefore, we state the next theorem in terms of unnormalized measures.

Theorem 7. *Let $\mathcal{Q}$ be a finite set of probability distributions over a discrete random variable X . Let $\tilde{P}$ be a positive measure over X , and let P be the corresponding normalized distribution. Let R be an arbitrary convex combination of the elements of $\mathcal{R} \subseteq \mathcal{Q}$. Then*

$$\mathbb{D}_C[\mathcal{Q} \parallel P] \leq \sup_{Q \in \mathcal{R}} \mathbb{F}[Q, \tilde{P}] - \inf_{Q \in \mathcal{Q}} \mathbb{F}[Q, \tilde{P}] + \mathbb{D}_{\text{KL}}[R \parallel P],$$

where

$$\mathbb{F}[Q, \tilde{P}] = \mathbb{H}[Q(X)] + \mathbb{E}_Q[\ln \tilde{P}(X)].$$

Proof. By Corollary 1.1, we have

$$\mathbb{D}_C[\mathcal{Q} \parallel P] \leq \mathbb{S}_{\mathcal{Q}}[P] - g(C).$$

where C is the central distribution of $\mathcal{Q}$ for X .

Since R is a convex combination of the elements of $\mathcal{R}$, Lemma 5 implies

$$g(R) \geq \inf_{Q \in \mathcal{R}} \mathbb{D}_{\text{KL}}[Q \parallel P] - \mathbb{D}_{\text{KL}}[R \parallel P].$$

By Theorem 2, C is a maximizer of g , and therefore

$$g(C) \geq g(R).$$

Combining these three inequalities yields

$$\mathbb{D}_C[\mathcal{Q} \parallel P] \leq \mathbb{S}_{\mathcal{Q}}[P] - \inf_{Q \in \mathcal{R}} \mathbb{D}_{\text{KL}}[Q \parallel P] + \mathbb{D}_{\text{KL}}[R \parallel P]. \quad (18)$$

Let $\tilde{P}(X) = ZP(X)$, where $Z > 0$ is a normalization constant. Then for any probability distribution Q ,

$$\mathbb{D}_{\text{KL}}[Q \parallel P] = \ln Z - \mathbb{F}[Q, \tilde{P}].$$

Since $\ln Z$ does not depend on Q , we have

$$\begin{aligned}\inf_{Q \in \mathcal{R}} \mathbb{D}_{\text{KL}} [Q \parallel P] &= \ln Z - \sup_{Q \in \mathcal{R}} \mathbb{F} [Q, \tilde{P}] \\ \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q \parallel P] &= \ln Z - \inf_{Q \in \mathcal{Q}} \mathbb{F} [Q, \tilde{P}],\end{aligned}$$

and the result follows from Equation 18. $\square$

To compute an approximate central distribution, Theorem 7 suggests constructing a positive measure $\tilde{P}$ for which $\mathbb{F} [Q, \tilde{P}]$ is approximately constant with respect to Q . We may choose this constant to be zero and seek

$$\mathbb{F} [Q, \tilde{P}] \approx 0,$$

for every $Q \in \mathcal{Q}$. In addition, the normalized distribution P should be close enough to a convex combination of the elements of $\mathcal{Q}$.

Since central distributions are defined through marginal distributions, their construction may be viewed as combining prior specification and Bayesian inference within a single optimization framework. Given n observations, one may perform inference using the central distribution of the corresponding finite sequence S_n , without separately specifying a prior distribution.

For sequences of random variables, however, central distributions are defined on the underlying infinite sequence. Consequently, the central distribution of a finite initial segment S_n should be viewed as an approximation to the corresponding marginal of the infinite-sequence central distribution. The question then becomes how accurate this approximation is.

One approach to computing an approximate central distribution is to represent $\ln \tilde{P}(X)$ as a linear combination of a suitable set of basis functions, analogous to the log-density representation used in exponential families:

$$\ln \tilde{P}(x) = - \sum_i \alpha_i b_i(x).$$

Substituting this into $\mathbb{F}$ gives

$$\mathbb{F} [Q, \tilde{P}] = \mathbb{H} [Q(X)] - \sum_i \alpha_i \mathbb{E}_Q [b_i(X)].$$

Therefore, enforcing $\mathbb{F} [Q, \tilde{P}] \approx 0$ amounts to approximating the entropy function by a linear combination of the feature expectations $\mathbb{E}_Q [b_i(X)]$.

Suppose that $\mathcal{Q}$ is indexed by a parameter Θ . We may then choose the coefficients α_i so that for every θ

$$\sum_i \alpha_i \mathbb{E}[b_i(X) \mid \theta] \approx \mathbb{H}[Q(X \mid \theta)],$$

and the resulting probability distribution is given by

$$P(x) \propto \exp \left\{ - \sum_i \alpha_i b_i(x) \right\}.$$

It should be noted that the quality of the approximation depends on how well $\tilde{P}$ can be represented as a nonnegative linear combination of distributions $Q_\theta \in \mathcal{Q}$. This can be assessed approximately using a sample set of parameter values $\mathcal{S}$, by finding nonnegative weights w_i such that

$$\tilde{P}(x) \approx \sum_i w_i Q(x \mid \theta_i),$$

where $\theta_i \in \mathcal{S}$.

The following example illustrates the approximation procedure in a simple binomial setting.

Example 2.4. Consider a binomial model with parameter θ . Let

$$S_n = (X_1, X_2, \dots, X_n)$$

denote a sample of n independent observations, where $X_i \sim \text{Bernoulli}(\theta)$. We use the basis functions

$$b_k(s_n) := \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^k.$$

These basis functions form polynomial features of the sufficient statistic. For a binomial model with success probability θ , we have

$$\mathbb{E}[b_k(S_n) \mid \theta] = \theta^k,$$

and

$$\mathbb{H}[Q(S_n \mid \theta)] = -n(\theta \ln \theta + (1 - \theta) \ln(1 - \theta)).$$

We consider the case $n = 7$ and approximate $C(S_7)$ using the procedure described above. Here, C denotes the marginal distribution induced by the central prior $\theta^{-\frac{1}{2}}(1 - \theta)^{-\frac{1}{2}}$. The parameter space is represented by a sample set $\mathcal{S}$ consisting of 30 values of θ .

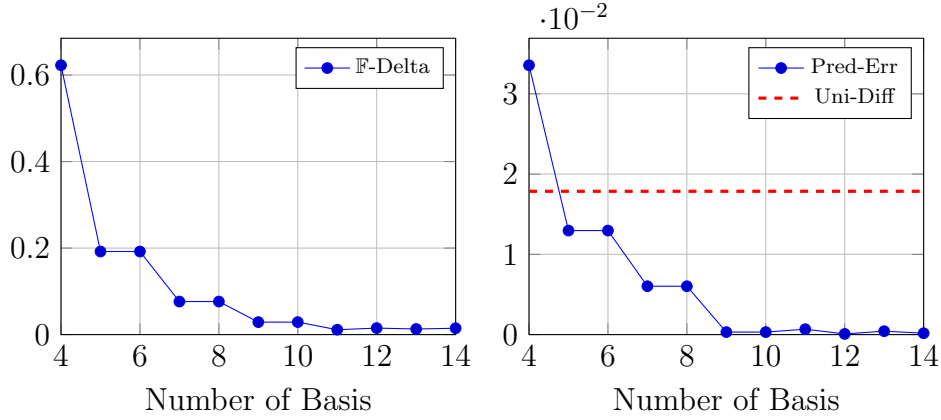


Figure 1: Approximation quality as a function of the number of basis functions. Left: $\mathbb{F}$ -Delta, the bound derived from Theorem 7. Right: Pred-Err the absolute difference between the estimated predictive probability and the central predictive probability. The dotted line (Uni-Diff) indicates the corresponding error obtained using a uniform prior.

When the approximation lies close to a marginal likelihood within the binomial family, the bound in Theorem 7 can be estimated by

$$\max_{\theta \in \mathcal{S}} \mathbb{F} \left[Q_{\theta}, \tilde{P} \right] - \min_{\theta \in \mathcal{S}} \mathbb{F} \left[Q_{\theta}, \tilde{P} \right],$$

which we denote by $\mathbb{F}$ -Delta.

To investigate the effect of the number of basis functions on the quality of the approximation, we vary the number of basis functions from 4 to 14. The corresponding values of $\mathbb{F}$ -Delta are shown in Figure 1.

Next, we use the approximation to estimate the central predictive probability $C(X_7 | S_6)$. We consider an observed sequence S_6 containing two successes and four failures. The corresponding predictive probability of success under C is $5/14$. Figure 1 reports the absolute difference between the estimated predictive probability and the central predictive probability, denoted by Pred-Err. For comparison, the absolute error of the predictive probability obtained under the uniform prior is shown as the dotted line (Uni-Diff).

The results show that as the number of basis functions increases, both $\mathbb{F}$ -Delta and Pred-Err decrease substantially. This suggests that improvements in the approximation bound are accompanied by improvements in the resulting predictive distribution. Moreover, with a sufficient number of basis functions, the predictive estimate becomes considerably closer to the central posterior than the estimate obtained from the uniform prior.

3 Illustrative Example: Attribute Selection

A natural application of hierarchical central distributions arises in feature selection. When the relevant set of features is unknown, each candidate feature set induces a distinct family of admissible distributions. Feature selection can therefore be viewed as uncertainty over a hierarchy of probabilistic models, making it a natural setting for the central distribution framework.

To illustrate this idea, we provide a proof of concept experiment using synthetic data. The purpose of this section is not to establish a competitive feature selection procedure, but to illustrate how hierarchical central distributions can be used to represent uncertainty over model structure. A comprehensive empirical evaluation against established feature selection methods, such as Spike-and-Slab priors, on real-world datasets lies beyond the scope of this foundational paper and is left for future work.

We consider a classification problem where the goal is to predict a target class, denoted by Y , from a set of observed nominal attributes $(A_1, A_2, \dots, A_n)$. We assume that Y may depend on an unknown subset of the attributes, rather than the full set.

To capture this relationship, we use the fused multinomial model introduced in Example 2.3. As an example, assume that Y depends only on A_2 and A_5 . Then, we define the labeling function f as follows:

$$f(A_1, A_2, \dots, A_n, Y) = (A_2, A_5, Y).$$

In a fused multinomial model with this labeling function, the target class Y is conditionally independent of all other attributes given A_2, A_5 .

For every possible subset of attributes, a distinct fused multinomial model of this form is constructed. Each such model corresponds to a set of plausible probability distributions. Collecting these sets yields our overall uncertainty representation, which forms a one-level hierarchy of distributions (Definition 2).

Let $C_{\mathcal{H}^*}$ denote a central distribution of the full hierarchy, and let C_{f_i} denote the central distribution of the fused multinomial model associated with the labeling function f_i . As discussed earlier, because the number of candidate models is finite, we have

$$C_{\mathcal{H}^*}(S_n) \propto \sum_i C_{f_i}(S_n). \quad (19)$$

To justify this, observe that we can construct a consistent estimator for the model indicator I by examining equality relationships among the observed frequencies of outcomes. Specifically, the estimator identifies outcomes

Table 1: Mappings of 3-bit inputs under class functions Y_1 and Y_2

Input (bits)	$Y_1(\text{Input})$	$Y_2(\text{Input})$
000	a	A
001	b	B
010	c	A
011	d	C
100	e	D
101	f	D
110	g	D
111	h	E

that occur with equal frequencies and returns the indicator corresponding to the model that supports those equalities.

The optimal Bayesian decision rule depends on the choice of loss function. In this example, we use the zero-one loss, under which the optimal strategy predicts the target class with the highest posterior probability. These posterior probabilities are computed using the central distribution of Equation 19. It turns out that each subset-specific fused multinomial model is assigned its corresponding Jeffreys’ prior. Consequently, the resulting predictive distribution can be interpreted as Bayesian model averaging over all candidate feature subsets, with each model equipped with its Jeffreys’ prior. The central distribution framework provides a minimax-KL interpretation for this aggregation.

To see how this approach works in practice, we carried out a synthetic experiment with the following setup. We considered binary attributes and generated each attribute independently with equal probability for 0 and 1. The target class was defined as a deterministic function of a fixed subset of 3 out of 12 total attributes, as shown in Table 1. The same subset of relevant attributes and class assignment function were used across all trials.

For each experiment, we generated a test set of size 100. A larger test set was used to reduce variability in performance estimates, taking advantage of the synthetic nature of the data. To assess robustness to irrelevant features, we progressively removed one attribute at a time from the full set of 12, starting with irrelevant ones and stopping just before removing any of the three relevant attributes.

At each step, we trained and tested two versions of the central distribution: one assuming that the target class is a deterministic function of the attributes (CD-Det), and one allowing the target class to follow a conditional probability distribution given the attributes (CD-Prob). Both versions are

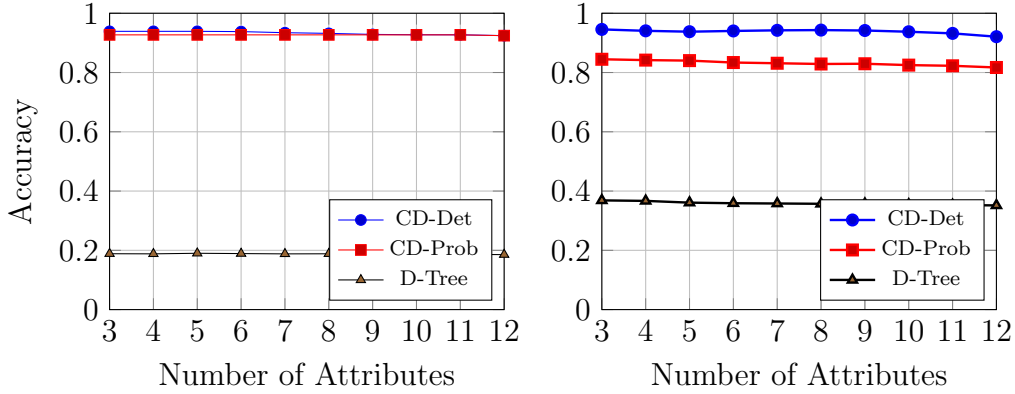


Figure 2: Accuracy versus number of attributes for 20 training samples. Left: under class assignment function Y_1 . Right: under class assignment function Y_2 .

constructed by aggregating over 2^{12} fused multinomial models, corresponding to all subsets of the 12 attributes. As a baseline, we also trained a decision tree classifier (D-Tree). All methods were trained on the same training data and evaluated on the same test data. Each configuration was repeated for 100 independent trials, and we reported the average accuracy (correct predictions divided by total test instances) over these trials.

The results, presented in Figure 2, show that both central distribution methods substantially outperform the decision tree on the two classification tasks. Interestingly, the relative difficulty of the tasks differs between the methods. For the central distribution approach, Y_2 is more challenging than Y_1 , resulting in a larger gap between the deterministic and probabilistic variants. This difficulty arises from strong non-deterministic dependencies between subsets of input bits and the assigned class, which can create misleading probabilistic associations. In contrast, the decision tree performs worse on Y_1 than on Y_2 . Overall, the deterministic central distribution achieved the highest accuracy, while the probabilistic version produced similar but slightly weaker results.

With 20 training samples, removing irrelevant attributes had little effect on performance. Accuracy remained nearly constant as the number of attributes decreased from 12 to 3, and the decision tree showed a similar insensitivity to attribute count. However, with only 10 training samples (Figure 3), accuracy increased noticeably as irrelevant attributes were removed, indicating that irrelevant features become more problematic in low-data regimes.

The implementation used in this experiment evaluates all 2^{12} candidate feature subsets and therefore remains computationally feasible only because

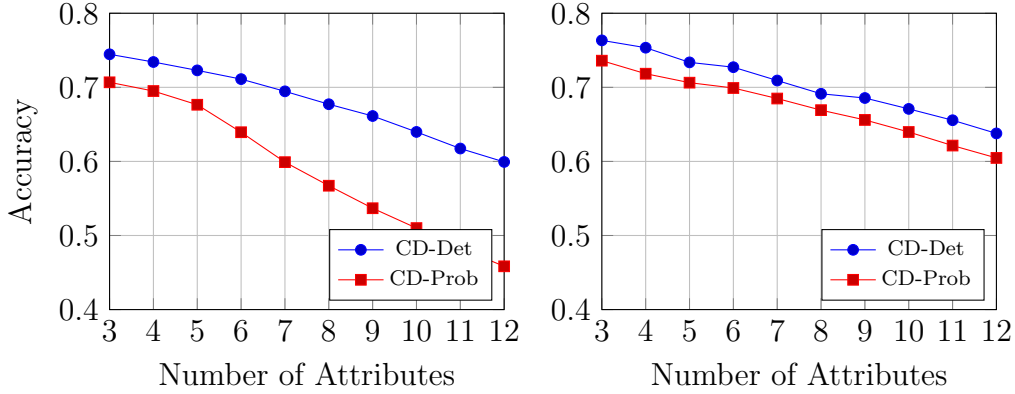


Figure 3: Accuracy versus number of attributes for 10 training samples. Left: under class assignment function Y_1 . Right: under class assignment function Y_2 .

the dimensionality is small. More generally, because the central distribution aggregates over all candidate subsets, the resulting predictor is not expected to scale to high-dimensional problems without additional approximation techniques. Developing scalable approximation methods could be a direction for future work.

4 Conclusion

We introduced the central distribution as a representative probability distribution for a set or hierarchy of admissible distributions. The central distribution is characterized by a minimax-KL principle, providing a principled probabilistic representation of partial knowledge while remaining within the standard framework of Bayesian probability theory.

For finite sets of distributions, we established existence, uniqueness, and characterization results and showed that central distributions admit a convex-combination representation. We extended the framework to hierarchies of distributions and introduced the notion of a central prior for parametric families. To address problems involving growing sequences of observations, we proposed the concept of asymptotic central distributions and derived sufficient conditions for asymptotic centrality. The resulting asymptotic framework suggests a connection between central priors and classical noninformative priors, including Jeffreys' prior in several standard models. In addition, we developed approximation bounds that lead to practical procedures for constructing approximate central distributions.

The framework suggests a new way of representing uncertainty about probabilistic models themselves. Rather than selecting a single model or beginning with a prior distribution over models, uncertainty may be represented directly through sets or hierarchies of admissible distributions and summarized by their corresponding central distributions. From this perspective, central distributions provide a bridge between robust, information-theoretic, and Bayesian approaches.

Several important questions remain open. In particular, a rigorous extension of the theory to general parametric families, more computational methods for computing and approximating central distributions, and a rigorous asymptotic theory for central priors require further investigation. It would also be valuable to study central distributions in a broader range of practical statistical problems in order to better understand their computational and inferential properties. We hope that the central distribution framework provides a useful foundation for studying noninformative priors and the representation of hierarchical partial knowledge from an information-theoretic perspective.

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